

# 복사전달 특강

## Special Topics in Radiative Transfer

### Lecture 6

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선광일 (Kwangil Seon)

KASI / UST

# Chapter 2

# Sampling Techniques

Random Number Generation · Inversion Methods · Rejection Sampling  
Normal Distribution · Algorithm Testing

## 2.1 Uniform Distribution & Pseudorandom Numbers

### Multiplicative Congruential Algorithm

$$x_{i+1} = (a \cdot x_i + c) \bmod m$$

$$\gamma_{i+1} = \frac{x_{i+1}}{m}$$

Parameters:

$a$  = multiplier

$m$  = modulus

$x_0$  = seed

$$a = 1103515245$$

$$c = 12345$$

$$m = 2^{31}$$

All three parameters are positive integers.

### Key Properties

- Generates  $\gamma \in (0,1)$
- Deterministic (reproducible with same seed)
- Finite period — modern generators have very long periods
- Seed can be set randomly (e.g. system clock)
- For further reading: Fishman (1996)

### Sample from Uniform(a, b)

Problem: generate  $\xi \sim \text{Uniform}(a, b)$

1. Generate  $\gamma$
2.  $\xi = (b - a)\gamma + a$

### Uniform Point in 3D Rectangular Box

Box:  $[x_a, x_b] \times [y_a, y_b] \times [z_a, z_b]$

Generate  $\gamma \rightarrow \xi = (x_b - x_a)\gamma + x_a$

Generate  $\gamma \rightarrow \eta = (y_b - y_a)\gamma + y_a$

Generate  $\gamma \rightarrow \zeta = (z_b - z_a)\gamma + z_a$

The algorithm uses three different and statistically independent  $\gamma$ s.

## 2.2 The Inversion Method for Continuous Distributions

**Problem:** Generate a random number  $\xi$  that has a continuous distribution  $f(x)$ .

**Groundwork:**

Find CDF of  $f(x)$ :  $F(x) = \int_{-\infty}^x f(x')dx'$

**Steps:**

1. Generate a uniform random number  $\gamma$ :  
 $\gamma \in (0,1)$
2. Solve  $\gamma = F(\xi)$  for  $\xi$   
 $\rightarrow \xi = F^{-1}(\gamma)$

**Proof**

**By definition,**

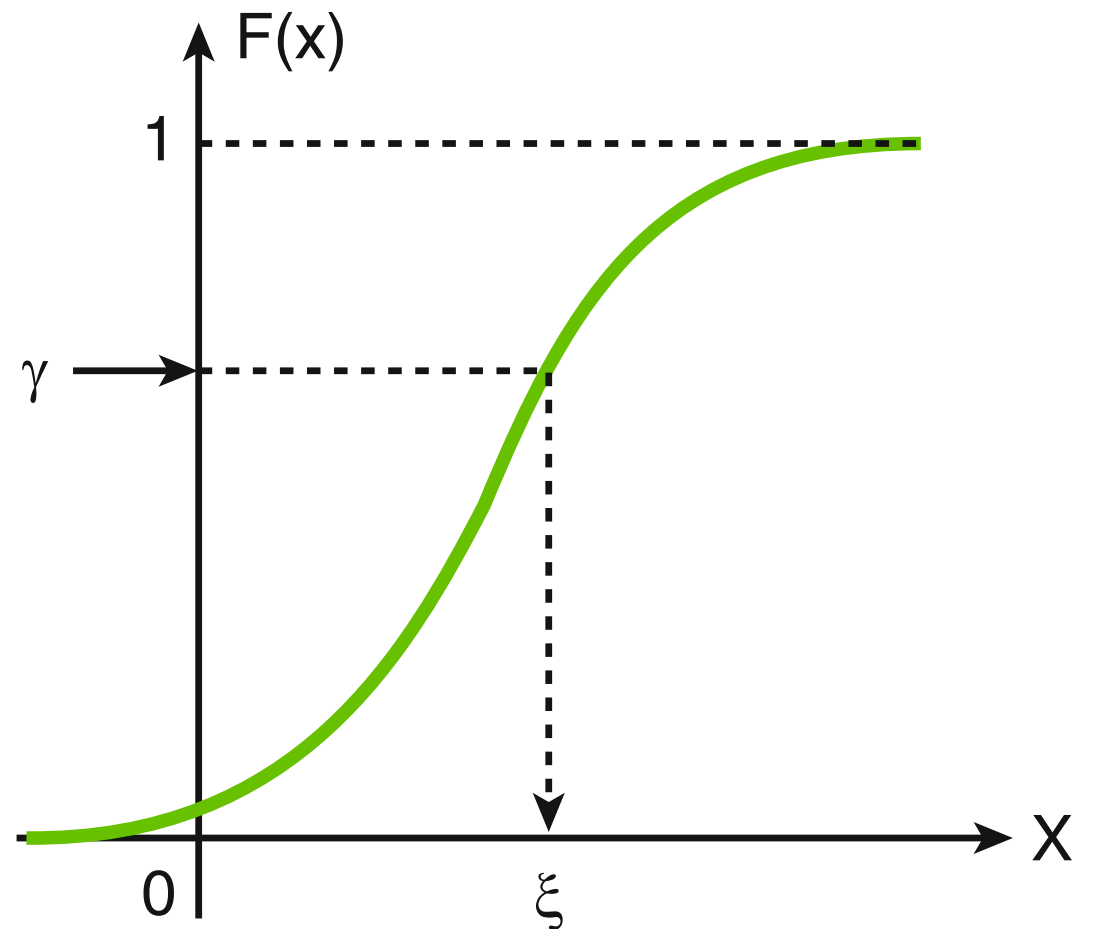
$$F(x) = P(\xi \leq x) = P\{F^{-1}(\gamma) \leq x\}$$

**Because  $F(x)$  is monotonic,**

$$\begin{aligned} &= P\{F[F^{-1}(\gamma)] \leq F(x)\} \\ &= P\{\gamma \leq F(x)\} \end{aligned}$$

**This proves that the algorithm samples  $\xi$  correctly.**

**CDF Inversion**



## 2.2

### Example Applications

#### Ex 1: Chord-Length Distribution

A parallel beam of particles is incident on a sphere of unit diameter. Each particle entering the sphere travels a random path  $\xi$  (chord) within it.

Assume that the particles travel along a straight line, and do not stop within the sphere, the chord-length distribution is given by this simple formula:

$$f(x) = \begin{cases} 2x, & \text{if } 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$F(x) = x^2 \quad \text{for } 0 \leq x \leq 1$$

**Algorithm:**

1. Generate  $\gamma$
2.  $\xi = \sqrt{\gamma}$

#### Ex 2: Free Path (Exponential)

Particles travel along a straight line between points of interaction with matter. The distance between interactions is called a free path. It is distributed exponentially:

$$f(x) = \begin{cases} \sigma \exp(-\sigma x), & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Here,  $\sigma$  is the optical depth per unit length ( $\text{cm}^{-1}$ ) and  $x$  is the distance (cm).

$$F(x) = 1 - \exp(-\sigma x)$$

**Algorithm:**

1. Generate  $\gamma$
2.  $\xi = -(1/\sigma) \ln \gamma$

## 2.3 The Inversion Method for Sampling Discrete Distributions

### 2.3 Discrete CDF Inversion

- For discrete random numbers  $\{x_1, x_2, \dots\}$ , the CDF is a stepwise-constant function
- Example — die roll:  
 $x_k = k$ , and  $f(k) = 1/6$  for  $k = 1, \dots, 6$   
 $F(k) = k/6$

#### Generic Algorithm:

- (1)  $k = 1$   
Sample  $\gamma$ ;  $\delta = \gamma$
- (2) Compute  $P_k$ ;  $\delta = \delta - P_k$
- (3) If  $\delta \leq 0 \rightarrow$  output  $\xi = x_k$ , exit  
Else  $k \Rightarrow k + 1$ ; repeat

#### Useful Recurrence Relations:

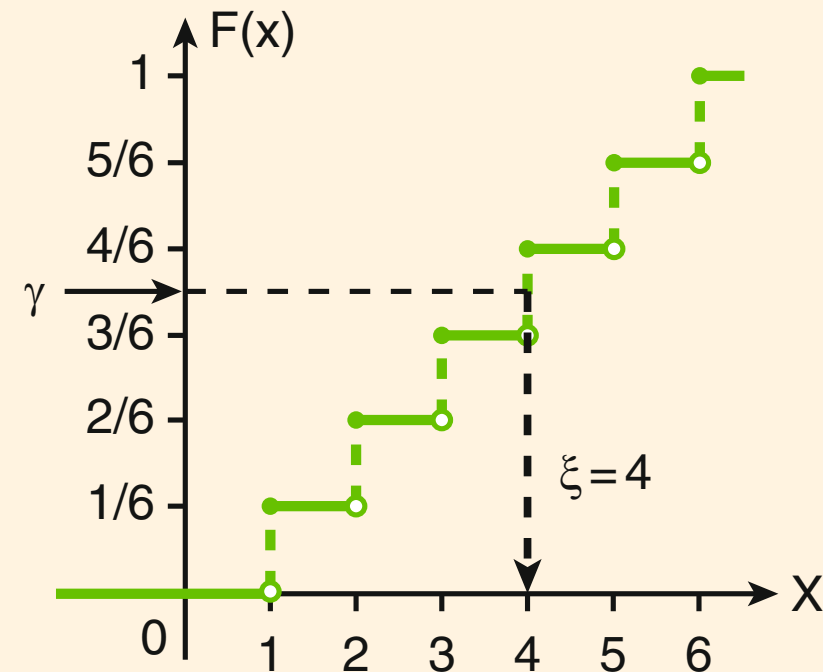
##### Binomial:

$$P_{k+1} = \frac{n-k}{k+1} \frac{p}{1-p} P_k; \quad P_0 = (1-p)^n; \quad k = 0, 1, \dots$$

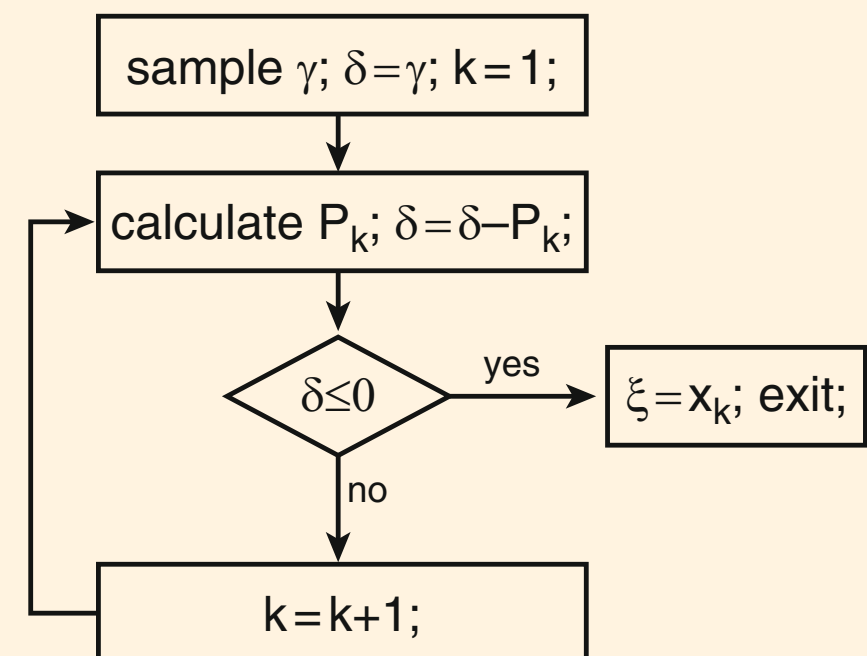
##### Poisson:

$$P_{k+1} = \frac{\mu}{k+1} P_k; \quad P_0 = \exp(-\mu); \quad k = 0, 1, \dots$$

### Inversion of the CDF for one roll of a die



#### Algorithm



## 2.4 Simple Rejection Method

The CDF inversion method is not always applicable to every distribution, nor is it necessarily the fastest or simplest method. The rejection method is a good alternative that offers a very straightforward algorithm.

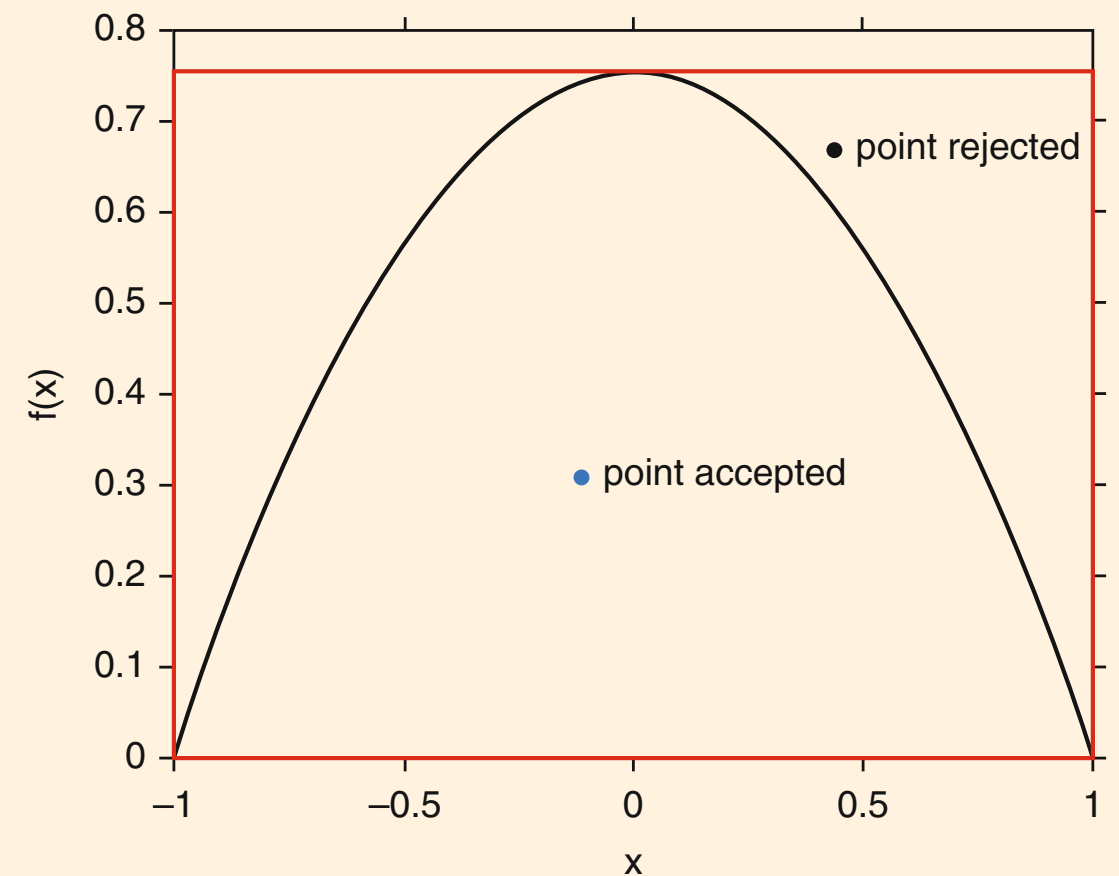
- Find minimum bounding box for  $f(x)$  (the red box)
- Place random point uniformly in box
- Accept if point falls below  $f(x)$  curve

**Algorithm:**

1. Sample  $\xi \sim \text{Uniform}[x_{\min}, x_{\max}]$
2. Sample  $\eta \sim \text{Uniform}[0, \max f(x)]$
3. If  $\eta < f(\xi) \rightarrow \text{ACCEPT } \xi$   
Else  $\rightarrow \text{REJECT, repeat}$

*The efficiency of the algorithm, defined as the probability of accepting a point, is*

*Efficiency = Area under  $f(x)$  / Box area*



**Example:**

$$f(x) = (3/4)(1 - x^2) \text{ for } -1 \leq x \leq 1$$

- (1) Sample  $\xi$  uniformly within  $[-1, 1]$ , and  $\eta$  uniformly within  $[0, 3/4]$ .
- (2) If  $\eta > (3/4)(1 - \xi^2)$ , then the point is rejected, go back to step 1.  
Else, the point is accepted; output  $\xi$ .

$$\text{Efficiency} = 2/3 \approx 66.7\%$$



## 2.5 Neumann's Method (Rejection with Proposal Distribution)

### Algorithm Setup

Choose proposal  $g(x)$  such that:

- (a)  $g(x)$  is similar to  $f(x)$
- (b)  $g(x)$  is easy to sample
- (c) A number  $C \geq 1$  exists such that  $Cg(x) \geq f(x)$  for all  $x$   
→ Choose  $g(x)$  with the smallest  $C$ .

### Algorithm Steps:

1. Sample  $\xi$  from  $g(x)$
2. Generate a uniformly-random number  $\gamma$
3. If  $\gamma \leq f(\xi)/[C \times g(\xi)]$   
→ output  $\xi$  and exit  
Else → go back to step 1

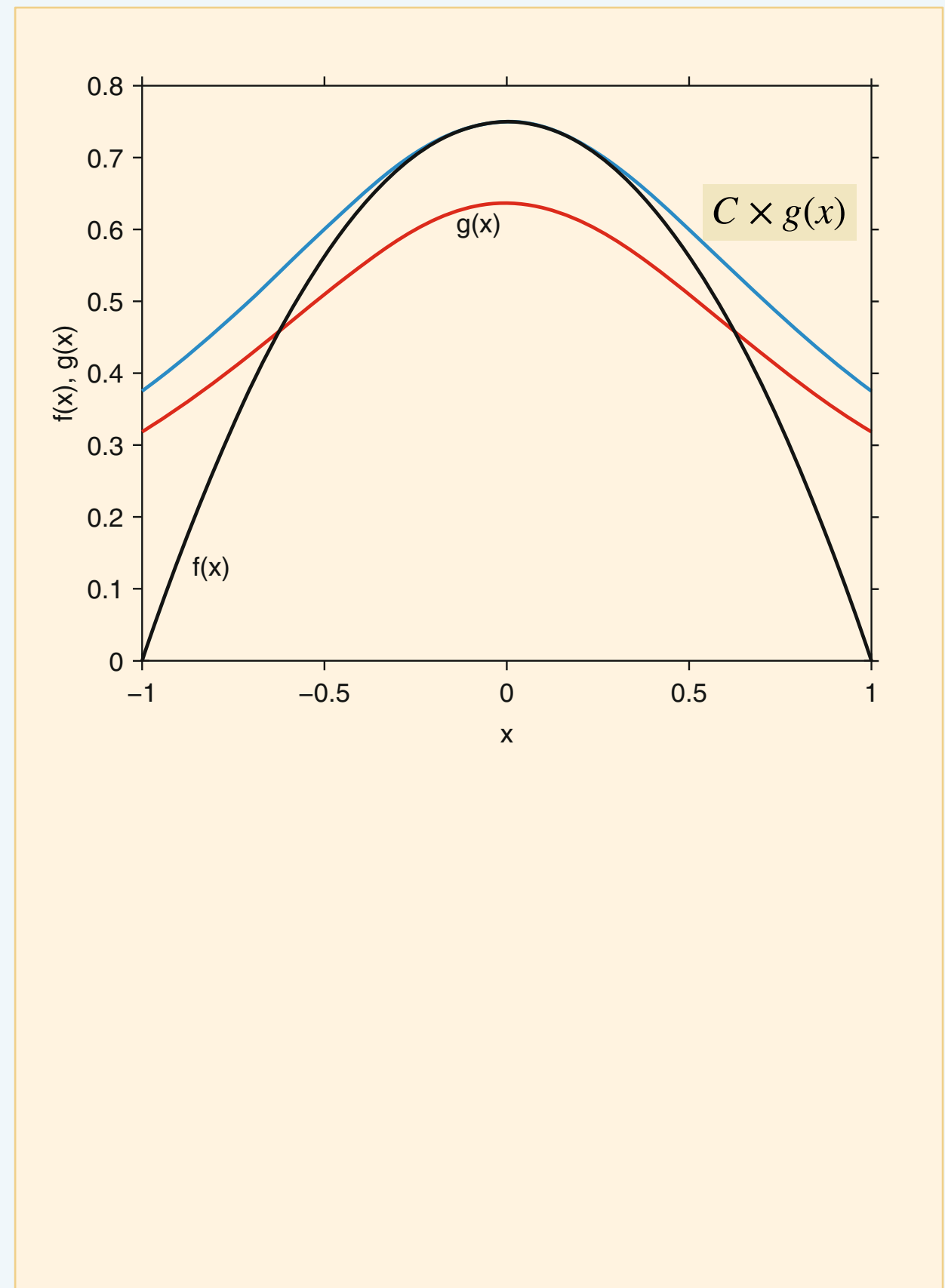
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$$\text{Efficiency} = \text{Area under } f(x) / \text{Area under } C \cdot g(x) \\ = 1/C$$

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If  $g(x) = \text{const} \rightarrow$  same as simple rejection

If  $g(x) = f(x) \rightarrow C=1$ , no rejection needed





## 2.5 Neumann's Method (Rejection with Proposal Distribution)

### Example Application

**Sample from**  $f(x) = (3/4)(1 - x^2)$ ,  $-1 \leq x \leq 1$

Choose proposal distribution:

$$g(x) = \frac{2}{\pi} \frac{1}{1 + x^2} \quad (\text{Cauchy})$$

$$C = 3\pi/8 \approx 1.178$$

*Efficiency  $\approx 85\%$*

*(vs 67% for simple rejection)*

**Sampling formula for  $g(x)$ :**

$$\xi = \tan \left[ \frac{\pi}{2} \left( \gamma - \frac{1}{2} \right) \right]$$

**Acceptance test:**

$$\gamma \leq f(x)/[Cg(x)] = 1 - \xi^4$$

*Note:  $\sin^2\vartheta$  cancels out elegantly in the acceptance condition for this specific  $f(x)$  and  $g(x)$  pair.*

## 2.6 Variable Transformation

It may be possible to transform a random variable  $\xi$  into another variable  $\eta = g(\xi)$  that has a distribution that is easier to sample.

Then, first sample  $\eta$  and then perform the inverse transformation  $\xi = g^{-1}(\eta)$ .

$$g_{\eta}(y) = f_{\xi}(x) \left| \frac{dx}{dy} \right| = \frac{f_{\xi}(x)}{\left| dg(x)/dx \right|}$$

The right-hand side should be expressed as a function of  $y$  using  $x = g^{-1}(y)$ .

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This method is also applicable to multidimensional variables.

$$g_{\eta_1, \dots, \eta_n}(y_1, \dots, y_n) = f_{\xi_1, \dots, \xi_n}(x_1, \dots, x_n) \left| \det \left\{ \frac{\partial x_i}{\partial y_j} \right\} \right|, \quad i, j = 1, 2, \dots, n$$

**Example:**

Sample distribution

$$f_{\xi}(x) = 2xe^{-x^2}, \quad x \geq 0$$

Instead, we sample  $\eta = \xi^2$ .

The distribution of  $\eta$  is

$$\rightarrow g_{\eta}(y) = f_{\xi}(x) \frac{1}{2x} = e^{-y} \quad (\text{exponential!})$$

**Algorithm:**

1. Sample  $\eta$  from  $\exp(-y)$  for  $y \geq 0$  via the CDF inversion method:

$$\eta = -\ln(\gamma)$$

2. Output  $\xi = \sqrt{\eta}$

## 2.7 Sum of Distributions

### Sample distribution

$$f(x) = \sum_i p_i f_i(x)$$

$$\text{where } p_i \geq 0, \sum_i p_i = 1, \text{ and } \int f_i(x) dx = 1$$

### Algorithm:

1. Sample a channel number  $i$  from the discrete distribution given by probabilities  $\{p_1, p_2, \dots\}$
2. Sample  $\xi$  from  $f_i(x)$

### Physical Example — Photon Interactions:

Sample states of all particles after the interaction of a photon with an atom. We have three channels:

1.  $p_1 \rightarrow$  Photoelectric absorption
  - the photon is absorbed
  - a photoelectron is ejected
  - the atom is ionized
2.  $p_2 \rightarrow$  Compton scattering
  - the photon momentum changes (deflected)
  - a Compton electron is ejected
  - the atom is ionized
3.  $p_3 \rightarrow$  Pair production
  - the photon disappears
  - an electron and a positron are produced.

*First, we sample interaction type (the channel number, 1, 2, or 3).*

*Once the type of interaction is determined, we sample the parameters of particles after the interaction using the distribution function for the selected interaction type.*

## 2.8 Compton Scattering — Combined Algorithm

Klein–Nishina formula:

Sample the scattering angle and energy of a photon after Compton scattering. We consider a simple case where the photon with an energy  $E$  interacts with a free electron at rest. The probability distribution of photon energy ( $E'$ ) after scattering is given by

$$f(E') = A \left( \frac{E'}{E} + \frac{E}{E'} - \sin^2 \theta \right) \text{ where } \cos \theta = 1 - \frac{1}{E'} + \frac{1}{E} \Leftrightarrow E' = \frac{E}{1 + E(1 - \cos \theta)},$$

where the energies are in the units of electron rest energy,  $E = \text{energy}/m_e c^2$ .

The minimum and maximum  $E'$  can be found by setting  $\theta = \pi$  and  $\theta = 0$ , respectively.  
 $E'_{\min} = E/(1 + 2E)$ ;  $E'_{\max} = E$

**Strategy (3 Methods Combined; Neumann's method + a sum of two distributions + CDF inversion)**

- Decompose  $f(E')$  into  $Cg(E')h(E')$ :

$$f(E') = A \left( \frac{E'}{E} + \frac{E}{E'} \right) \left( 1 - \frac{\sin^2 \theta}{E'/E + E/E'} \right)$$

$$g(E') \propto \frac{E'}{E} + \frac{E}{E'} \quad \text{and} \quad h(E') = \left( 1 - \frac{\sin^2 \theta}{E'/E + E/E'} \right) \leq 1$$

$$\Rightarrow f(E') \leq A \left( \frac{E'}{E} + \frac{E}{E'} \right) = Cg(E')$$

## 2.8

- Sample the proposal distribution  $g(E')$ :

$$Cg(E') = A \left( \frac{E'}{E} + \frac{E}{E'} \right)$$

This function can be written as

$$Cg(E') = A(\alpha_1 + \alpha_2) \left[ \frac{\alpha_1}{\alpha_1 + \alpha_2} f_1(E') + \frac{\alpha_2}{\alpha_1 + \alpha_2} f_2(E') \right],$$

where

$$f_1(E') = A_1 E' \text{ and } f_2(E') = \frac{A_2}{E'}, \text{ and}$$

their normalization constants  $A_1$  and  $A_2$  are obtained from  $A_1 \int_{E'_{\min}}^{E'_{\max}} E' dE' = 1$  and  $A_2 \int_{E'_{\min}}^{E'_{\max}} \frac{dE'}{E'} = 1$ .

Then, it can be determined that

$$\alpha_1 = \frac{1}{A_1 E} \text{ and } \alpha_2 = \frac{E}{A_2}$$

Now, we choose

$$g(E') = \frac{\alpha_1}{\alpha_1 + \alpha_2} f_1(E') + \frac{\alpha_2}{\alpha_1 + \alpha_2} f_2(E'), \quad \text{and} \quad C = A(\alpha_1 + \alpha_2).$$

- Sample  $f_1(E')$ :

$$\xi = \sqrt{(E'_{\min})^2 + 2\gamma/A_1} \quad \text{using a uniform random number } \gamma$$

- Sample  $f_2(E')$ :

$$\xi = E'_{\min} \exp(\gamma/A_2) \quad \text{using a uniform random number } \gamma$$

## 2.8

### Full Algorithm Steps

1. Generate  $\gamma$
2. If  $\gamma < \alpha_1/(\alpha_1 + \alpha_2)$  then sample  $\xi$  from  $f_1(E')$ :  
Generate (a new)  $\gamma$   
$$\xi = \sqrt{(E'_{\min})^2 + 2\gamma/A_1}$$
Else, sample  $\xi$  from  $f_2(E')$   
Generate (a new)  $\gamma$   
$$\xi = E'_{\min} \exp(\gamma/A_2)$$
3. Calculate  $\sin^2 \theta = 1 - \left(1 - \frac{1}{\xi} + \frac{1}{E}\right)^2$  from  
Compton energy and momentum conservation  
formula
4. Generate  $\gamma$
5. If  $\gamma < 1 - \sin^2 \theta / (\xi/E + E/\xi)$   
→ output  $E' = \xi$  and  $\theta$ ; exit  
Else, go back to step 1

## 2.9 Superposition Method

Sample a probability distribution defined as an integral

$$f(x) = \int h(x, t) dt$$

Choose a normalized and simple-to-sample distribution  $g(t)$ , and rewrite the integral as

$$\int h(x, t) dt = \int g(t) \frac{h(x, t)}{g(t)} dt \Rightarrow \int g(t) f(x|t) dt$$

The ratio  $h/g$  can be interpreted as a conditional probability distribution. Hence, the algorithm is:

**Algorithm:**

1. Sample  $\tau$  from distribution  $g(t)$
2. Sample  $\xi$  from distribution  $f(x|\tau)$

**Example**

Sample the following distribution:

$$f(x) = A \int_0^\infty t \exp[-t(x+t)] dt \quad (x \geq 0 \text{ and } A > 0)$$

Normalization factor:

$$\int_0^\infty f(x) dx = 1 \rightarrow A = \frac{2}{\sqrt{\pi}}$$

The integrand can be written as a product of two normalized distributions:

$$At \exp[-t(x+t)] = t \exp(-tx) \cdot \frac{2}{\sqrt{\pi}} \exp(-t^2)$$

**Algorithm:**

1. Sample  $\tau$  from  $(2/\sqrt{\pi})\exp(-t^2)$ ,  $t \geq 0$ .  
This is the positive half of a normal distribution.
2. Sample  $\xi$  from  $\tau \exp(-\tau x)$ ,  $x \geq 0$ . This is the exponential distribution.



## 2.10 Sampling the Normal Distribution $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[ -\frac{(x - \mu)^2}{2\sigma^2} \right] \quad \text{Approach: sample } \delta \sim N(0,1), \text{ then output } \xi = \sigma\delta + \mu$$

### Method 1: Central Limit Theorem Approximation

1. Generate  $\gamma_1, \gamma_2, \dots, \gamma_n$
2. Sum the  $n$  uniform random numbers:

$$\xi = \sqrt{\frac{12}{n}} \sum_{i=1}^n \left( \gamma_i - \frac{1}{2} \right)$$

$n = 12$  has been recommended.

$$\xi = \left( \sum_{i=1}^{12} \gamma_i \right) - 6$$

It is easy to show that  $\text{Var}(\xi) = 1$  and  $E(\xi) = 0$ .

⚠ Inaccurate in distribution tails

### Method 2: Box-Muller Transform

Produces 2 independent  $N(0,1)$ :

1. Generate  $\gamma_1$  and  $\gamma_2$
2. Calculate  $\phi$  and  $\rho$ :  
 $\phi = 2\pi\gamma_1$   
 $\rho = \sqrt{-2 \ln \gamma_2}$

3. Output two samples:  
 $\xi_1 = \rho \cos \phi$   
 $\xi_2 = \rho \sin \phi$

Exact – but requires sin/cos.

### Method 3: Geometric Rejection

Avoid sin/cos computation:

1. Generate  $\gamma_1$  and  $\gamma_2$
2. Sample  $(\eta_1, \eta_2)$  in unit circle:  
 $\eta_1 = 2\gamma_1 - 1$  and  $\eta_2 = 2\gamma_2 - 1$
3. If  $\alpha^2 = \eta_1^2 + \eta_2^2 > 1$ , reject
4. Else the point is within the circle.
5. Generate  $\gamma_3$
6. Calculate  $\rho = \sqrt{-2 \ln \gamma_3}$
7. Find  $x$  and  $y$  positions:  
 $\xi_1 = \eta_1 \rho / \alpha$  and  $\xi_2 = \eta_2 \rho / \alpha$

Efficiency =  $\pi/4 \approx 78.5\%$



## 2.11 Random Points & Directions

### Uniform Point within a Circle (R)

[A] Rejection:

1. Generate  $\gamma_1$  and  $\gamma_2$
2.  $\xi_1 = R(2\gamma_1 - 1)$ ,  $\xi_2 = R(2\gamma_2 - 1)$ , where  $R$  is the radius of the circle.
3. If  $x_1^2 + \xi_2^2 > R^2$ , the the point is outside the circle; go to step 1.
4. Else, output coordinates  $(\xi_1, \xi_2)$  of the point

[B] CDF Inversion (in polar coordinates):

Polar coordinates distribution:

$$f(\rho, R) = \frac{\rho}{\pi R^2}, \quad \rho < R$$

1. Generate  $\gamma_1 \rightarrow \phi = 2\pi\gamma_1$
2. Generate  $\gamma_2 \rightarrow \rho = R\sqrt{\gamma_2}$
3. Calculate  $x$  and  $y$  coordinates of the point

$$\xi_1 = \rho \cos \phi$$

$$\xi_2 = \rho \sin \phi$$

### Uniform Point in Sphere (R)

In Cartesian coordinates:

$$f(x, y, z) = \frac{1}{(4\pi/3)R^3} \quad \text{if } x^2 + y^2 + z^2 < R^2$$

In spherical coordinates:

$$f(r, \theta, \phi) = \frac{3r^2}{R^3} \frac{\sin \theta}{2} \frac{1}{2\pi} \quad \text{if } r < R$$

$r$ ,  $\theta$ , and  $\phi$  are independent:

$$f(r) = (3r^2)/R^3$$

$$f(\theta) = (\sin \theta)/2 \quad \text{or} \quad f(\mu) = \frac{1}{2} \quad \text{for } \mu = \cos \theta$$

$$f(\phi) = 1/(2\pi)$$

1. Generate  $\gamma \rightarrow r = R \cdot \gamma^{1/3}$
2. Generate  $\gamma \rightarrow \mu = 2\gamma - 1$
3. Generate  $\gamma \rightarrow \phi = 2\pi\gamma$
4.  $(\xi_1, \xi_2, \xi_3) = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$

## 2.11

### Isotropic Direction $\hat{\Omega}$

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Consider a unit direction vector  $\vec{\Omega} = (\Omega_x, \Omega_y, \Omega_z)$ .

Isotropic means that, for any solid angle  $d\omega$ ,

$$P(\vec{\Omega} \in d\omega) = \frac{d\omega}{4\pi}$$

Since an infinitesimal solid angle is

$$d\omega = \sin\theta d\theta d\phi = d\mu d\phi,$$

$\mu$  and  $\phi$  are independent and uniformly distributed:

$$\mu \in [-1, 1] \text{ uniform}$$

$$\phi \in [0, 2\pi] \text{ uniform}$$

**Algorithm:**

1. Generate  $\gamma \rightarrow \mu = 2\gamma - 1$

2. Generate  $\gamma \rightarrow \phi = 2\pi\gamma$

The, the components are:

$$\Omega_x = (1 - \mu^2)^{1/2} \cos \phi$$

$$\Omega_y = (1 - \mu^2)^{1/2} \sin \phi$$

$$\Omega_z = \mu$$

### Uniform Random Points on a Spherical Surface

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Sample an isotropic vector  $\vec{\Omega} = (\Omega_x, \Omega_y, \Omega_z)$ .

Calculate  $x$ ,  $y$ , and  $z$  coordinates of the point:

$$\xi_1 = \Omega_x R$$

$$\xi_2 = \Omega_y R$$

$$\xi_3 = \Omega_z R$$

Here,  $R$  is the radius of the sphere.

## 2.12 Joint Distributions

Sample two random numbers that are not statistically independent.

**Joint PDF factorization:**

$$f_{\xi,\eta}(x, y) = f_{\xi}(x)f_{\eta}(y | \xi = x)$$

Marginal:  $f_{\xi}(x) = \int f_{\xi,\eta}(x, y)dy$

Conditional:  $f_{\eta}(y | \xi = x) = \frac{f_{\xi,\eta}(x, y)}{f_{\xi}(x)}$

**Algorithm:**

1. Sample  $\xi$  from marginal  $f_{\xi}(x)$
2. Sample  $\eta$  from conditional  $f_{\eta}(y | \xi = x)$

**Example — ionization:**

Sample the energy of the ejected electron first,  
then sample ejection angle  
(using energy-dependent conditional distribution).

## 2.13 Particle-Scattering Event

- 1. Sample  $E'$  (post-scatter energy)
- 2. Sample  $\theta$  (polar angle)
- 3. Sample  $\phi \sim \text{Uniform}[0, 2\pi]$
- 4. Rotate direction:  $\Omega \rightarrow \Omega'$

**Direction rotation (general  $\Omega$ ):**

$$\begin{aligned}\Omega'_x &= \Omega_x \cdot \cos\theta - \sin\theta \cdot \\ &\quad [\Omega_x \cdot \Omega_z \cdot \cos\phi + \Omega_y \cdot \sin\phi] / \sqrt{1 - \Omega_z^2} \\ \Omega'_y &= \Omega_y \cdot \cos\theta - \sin\theta \cdot \\ &\quad [\Omega_y \cdot \Omega_z \cdot \cos\phi - \Omega_x \cdot \sin\phi] / \sqrt{1 - \Omega_z^2} \\ \Omega'_z &= \Omega_z \cdot \cos\theta + \sqrt{1 - \Omega_z^2} \cdot \sin\theta \cdot \cos\phi\end{aligned}$$

*Tip: sample  $\sin\phi$  and  $\cos\phi$  directly via geometric rejection — avoids computing inverse trig.*

## 2.13 Simulating a Particle-Scattering Event

### Algorithm

1. Sample  $E'$  (post-scatter energy)
2. Sample the scattering angle  $\theta$  (polar angle)
3. Sample the azimuthal angle  $\phi \sim \text{Uniform}[0, 2\pi]$
4. Rotate direction:  $\vec{\Omega} \rightarrow \vec{\Omega}'$

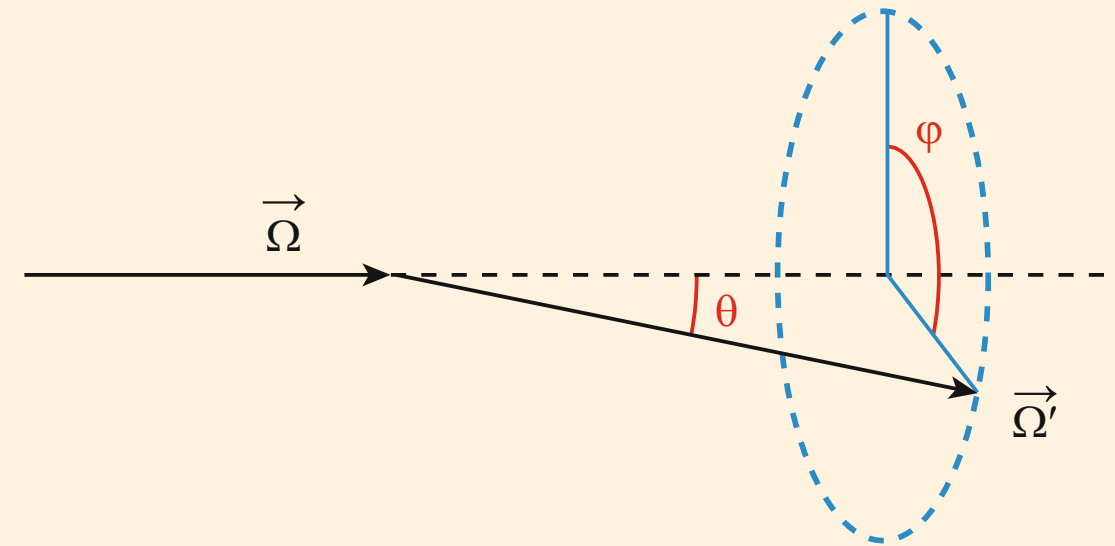
### Direction rotation (general $\Omega$ ):

$$\Omega'_x = \Omega_x \cos \theta - \frac{\sin \theta}{(1 - \Omega_z^2)^{1/2}} \left( \Omega_x \Omega_z \cos \phi + \Omega_y \sin \phi \right)$$

$$\Omega'_y = \Omega_y \cos \theta - \frac{\sin \theta}{(1 - \Omega_z^2)^{1/2}} \left( \Omega_y \Omega_z \cos \phi - \Omega_x \sin \phi \right)$$

$$\Omega'_z = \Omega_z \cos \theta - (1 - \Omega_z^2)^{1/2} \sin \theta \sin \phi$$

*Tip: sample  $\sin \phi$  and  $\cos \phi$  directly via geometric rejection — avoids computing inverse trig.*



$\vec{\Omega}$  = particle direction before the collision

$\vec{\Omega}'$  = particle direction after the collision

## 2.14 Algorithm Testing Methods

1

### Histogram (Qualitative)

1. Generate large sample  $\{\gamma_1, \gamma_2, \dots, \gamma_N\}$
2. Build normalized histogram:

$$h_i = \frac{n_i}{N\Delta x_i} = \text{height of the bar}$$

$\Delta x_i$  = histogram bin width

$n_i$  = number of random values that fell within the  $i$ -th bin.

$$N = \sum_i n_i = \text{sample size}$$

The histogram is correctly normalized:

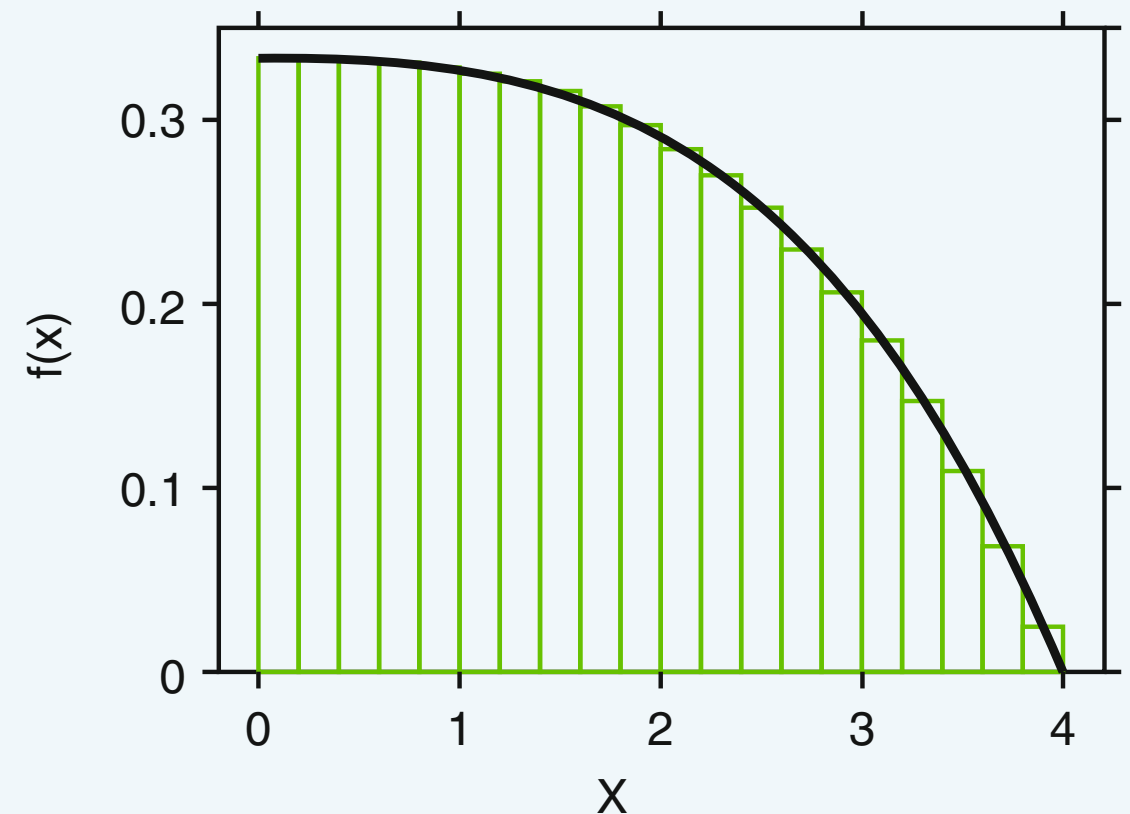
$$\sum_i h_i \Delta x_i = 1$$

3. Plot  $h_i$  vs  $f(x)$

→  $f(x)$  should pass through the centers of the histogram bars at their tops.

⚠ Narrow bin at singularities

⚠ If  $f(x)$  has a discontinuity, a bin boundary should be placed exactly at the point of discontinuity.



## 2.14

### 2

#### $\chi^2$ Test (Quantitative)

1. Divide axis into  $k$  intervals

2. Expected probabilities:  $p_i = \int_{a_i}^{b_i} f(x)dx$

3. Observed probabilities:  $\pi_i = \frac{n_i}{N}$

4. Compute:

$$\chi^2(\text{obs}) = \sum_{i=1}^k \frac{(\pi_i - p_i)^2}{p_i}$$

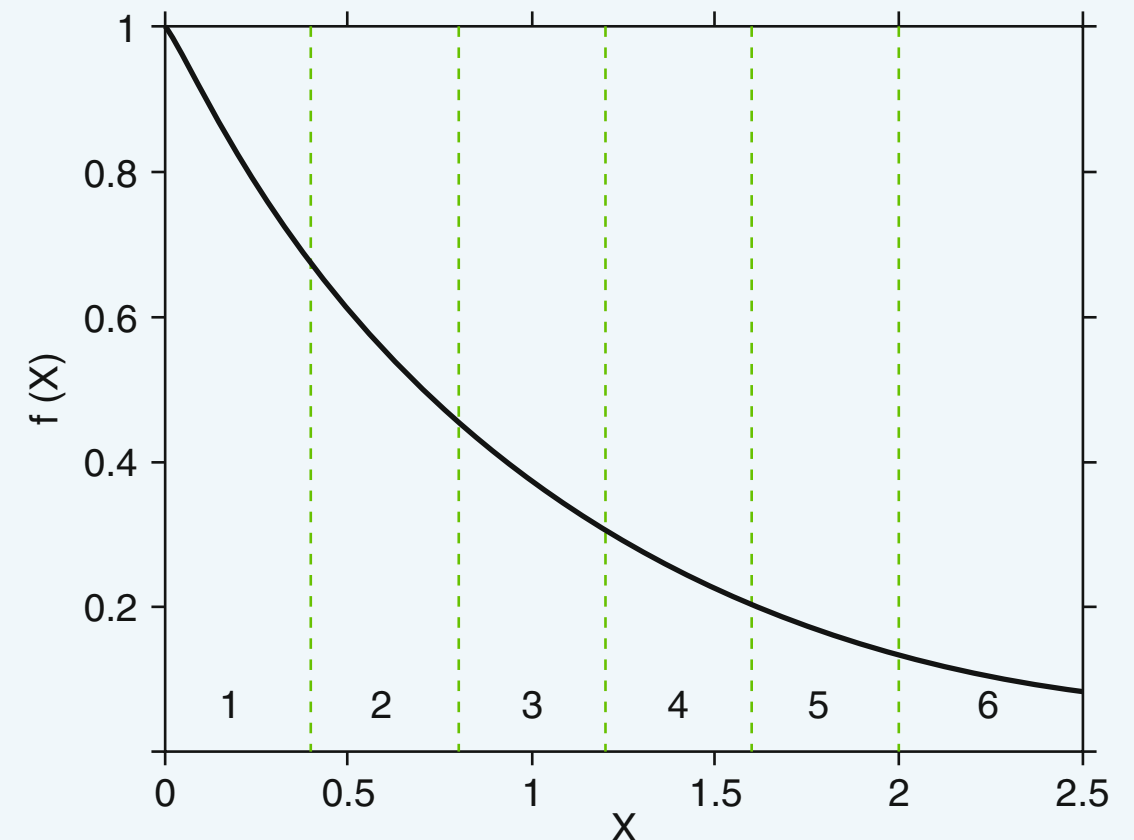
The sum is  $\chi^2$ -distributed if each  $n_i$  is much greater than 1 and  $k$  is sufficiently large.

5. Using the  $\chi^2$  distribution, calculate the probability of observing a  $\chi^2$  that is equal or greater than  $\chi^2(\text{obs})$ , using

$$P\{\chi^2 \geq \chi^2(\text{obs})\} = \int_{\chi^2(\text{obs})}^{\infty} p_{\chi^2}(x)dx$$

6. If the above probability is small, the hypothesis is rejected.

For example,  $P(\chi^2 \geq \chi^2(\text{obs})) < 0.05$  or  $0.001$ .



## 2.14

### 3

#### Likelihood Ratio (Sensitive)

$\theta_0$  = exact parameters

$\hat{\theta}$  = the maximum likelihood estimates of  $\theta$

1. Generate a large sample  $\{\xi_1, \xi_2, \dots, \xi_N\}$ .
2. Estimate parameters  $\hat{\theta}$  from sample
3. Compute ratio:

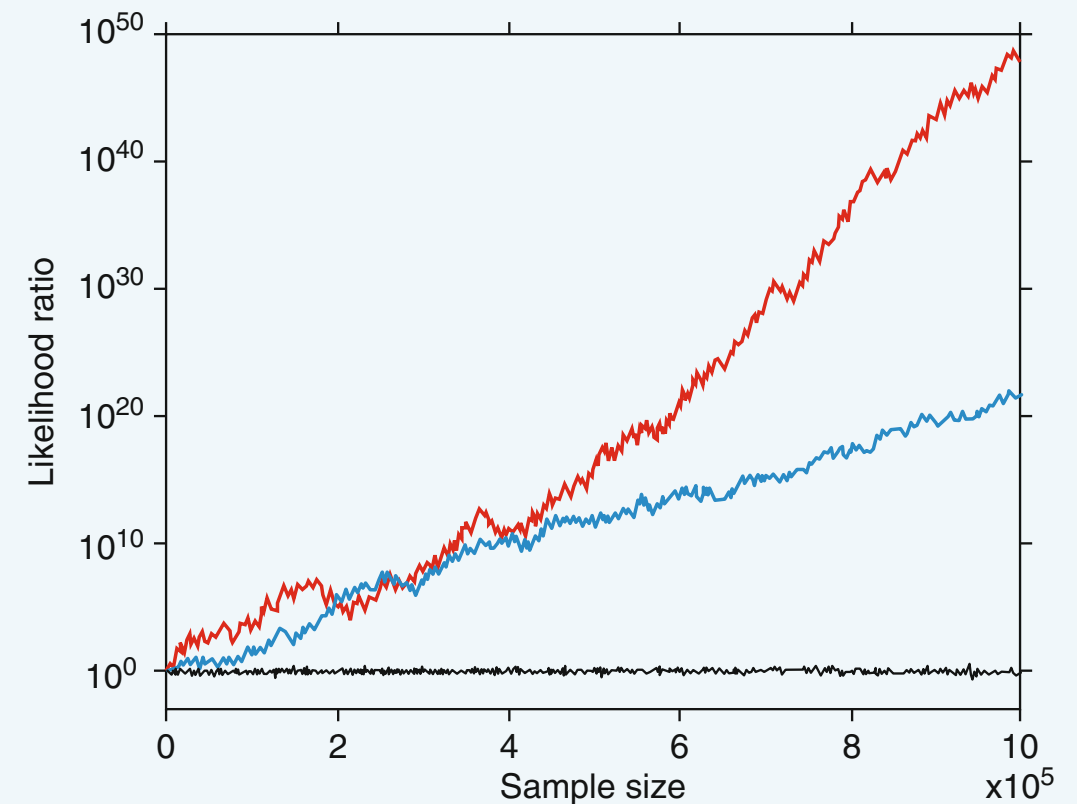
$$\lambda_i = \prod_{j=1}^i \frac{f(\xi_j | \theta_0)}{f(\xi_j | \hat{\theta})}; \quad i = 1, 2, \dots, N$$

4. Plot  $\lambda_i$  vs  $i$

Correct code  $\rightarrow \lambda_i$  will fluctuate around 1 (stable)

Error  $\rightarrow \lambda_i$  diverges.

Very sensitive – detects even 1% errors in parameters!



Likelihood ratio test for a code sampling normal distribution  $N(1,1)$ .

The black line shows the likelihood for a correct code.

The blue line indicates the code with a 1% error in the mean.

The red line represents a 1% error in the standard deviation.